

Dora Rajonhson, PhD

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PROFILE

Interdisciplinary scientist working at the interface of **bioinformatics, microbiome ecology, and public health**, with a long-term objective to understand how **host–microbiome–environment interactions shape biological resilience to environmental stressors** across systems. My research is driven by **cross-disciplinary, high-risk questions** that cannot be addressed within a single model organism or methodological framework. Trained across medical entomology, microbial ecology, and multi-omics, I have developed and led integrative analytical approaches spanning insects, aquatic organisms, and environmental samples. I aim to establish an **independent research program** that bridges ecological theory and omics technologies to address global challenges in health and sustainability.

EDUCATION

PhD in Tropical Medicine — Mahidol University, Thailand | 2017–2022

Thesis: *Integrative Morphometric, Molecular, and Microbial Characterization of Cimicidae Ectoparasites for Improved Identification and Public Health Surveillance*

MSc in Biology — University of Antananarivo, Madagascar | 2013–2015

Thesis: *Evaluation of Systemic Fipronil Application for the Control of the Plague Vector *Xenopsylla cheopis* in Madagascar*

RESEARCH EXPERIENCE

Program Management Unit for Human Resources & Institutional Development, Research and Innovation (PMUC)-Industrial Postdoctoral Fellow — National Center for Genetic Engineering and Biotechnology (BIOTEC), Thailand | 2024–2025

Integrated Transcriptomic and Microbial Profiling of Penaeid Shrimp under Environmental Stress

- Designed and led a **longitudinal multi-omics framework** integrating gut microbiome dynamics and transcriptomics to assess sublethal mycotoxin exposure over a 13-week culture period.
- Generated the **first multi-mycotoxin transcriptomic dataset in shrimp**, linking molecular responses to environmental modulation.

Postdoctoral Fellow (BIOTEC–UNESCO–TWAS) — BIOTEC, Thailand | 2023–2024

Impacts of Probiotic Treatment and Sequencing Strategies on Aquaculture Microbiomes

- Conceptualized comparative sequencing benchmarks combining short- and full-length 16S

rRNA approaches.

- Developed analytical pipelines to disentangle host-associated and environmental microbiome responses to probiotic intervention.

PhD Research Fellow (The Organisation for Women in Science for the Developing World (OWSD) — Mahidol University, Thailand | 2020–2022

- Integrated **morphometrics, molecular taxonomy, and microbiome profiling** to improve identification of medically relevant ectoparasites.
- Highlighted overlooked microbial reservoirs with potential zoonotic relevance at the human–wildlife interface.

MSc Researcher — Institut Pasteur de Madagascar | 2014–2015

- Conducted field and laboratory evaluations of systemic insecticide strategies targeting plague vectors in endemic settings.

AWARDS & FELLOWSHIPS

- PMUC–Industrial Postdoctoral Fellowship, Thailand (2024–2025)
- BIOTEC–UNESCO–TWAS Postdoctoral Fellowship, Thailand (2023–2024)
- OWSD–UNESCO PhD Fellowship, Thailand (2017–2022)
- Tranakchit Harinasuta Award — Most Outstanding PhD Student (2018)
- Gérard Scholarship for Master’s Thesis, Institut Pasteur de Madagascar (2014–2015)

SELECTED PUBLICATIONS

Peer-reviewed

Rajonhson DM et al. (2024). *Integrating short- and full-length 16S rRNA gene sequencing to elucidate microbiome profiles in Pacific white shrimp ponds*. **Microbiology Spectrum**, 12(11), e00965-24.

Rajonhson DM et al. (2022). *Comparison of two morphological methods to study the pronotum of Cimicidae*. **Insects**, 13(12), 1155.

Rajonhson DM et al. (2017). *Effectiveness of fipronil as a systemic control agent against *Xenopsylla cheopis**. **Journal of Medical Entomology**, 54(2), 411–417.

Manuscripts under review

Rajonhson DM et al. (2025). *Temporal dynamics and environmental drivers of gut and water microbiomes in *Litopenaeus vannamei* under zero-water exchange conditions*. **Environmental Microbiology**.

Rajonhson DM et al. (2025). *Co-occurrence of mycotoxins in aquaculture feed: transcriptomic insights into shrimp health*. **Comparative Biochemistry and Physiology D**.

TEACHING & SUPERVISION

PhD Co-supervisor — University of Antananarivo, Madagascar | Dec 2025–present

- Annick Raveloson — *Kdr gene amplicon sequencing and phylogeographic analysis*

MSc Supervisor — Thammasat University, Thailand | 2024–2025

- Ramon Akkharapreechanont — Fermented food microbiome project

Sessional Lecturer & Assistant — University of Antananarivo, Madagascar | 2016–2017

- Agricultural and medical entomology (lectures, tutorials, practical supervision)
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INVITED TALKS & INTERNATIONAL CONFERENCES

Invited Speaker

Journée Scientifique de l'Entomologie (1st edition) — Department of Entomology, Faculty of Science, University of Antananarivo, Madagascar

- *Integrating bioinformatics into medical entomology*

Selected International Conferences

- American Society of Tropical Medicine and Hygiene (ASTMH) Annual Meeting — 2025
 - 30th FAOBMB & 8th BMB Conference — 2023
 - Joint International Tropical Medicine Meeting — 2017
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TRAINING & WORKSHOPS (SELECTED)

- Viral Genomics and Bioinformatics — Wellcome Connecting Science, Oxford University Clinical Research Unit (OUCRU), Vietnam (2024)
 - Summer Institute in Statistics and Modeling in Infectious Diseases (SISMID) — University of Washington (2020)
 - International Short Course on Ecosystem Health — Thailand One Health University Network (2018)
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COMMUNITY & LEADERSHIP

Asia Chapter Coordinator (2021–2025) — Ikala STEM Madagascar (*association of women in STEM in Madagascar*)

- Coordinated regional initiatives promoting women's participation in STEM.
 - Acted as liaison between Asia-based members and global leadership.
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LANGUAGES

Malagasy (native) · English (fluent) · French (fluent)

REFERENCES

Available upon request